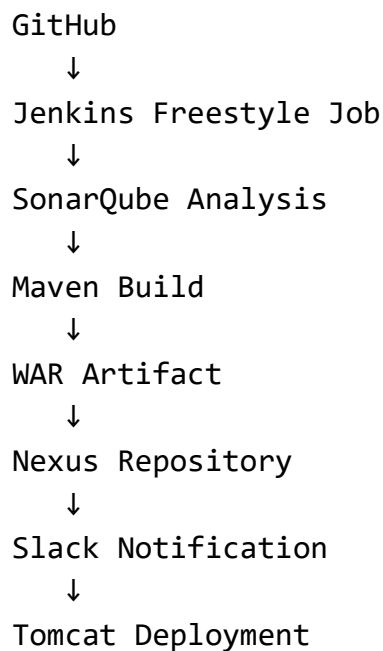


EOD – Jenkins Freestyle CI/CD Pipeline

1. Objective

Created a Jenkins **Freestyle CI/CD job** to automate the complete deployment process of a Java web application.

CI/CD Flow



2. Environment

Component	Configuration
Jenkins	2.584
OS	Amazon Linux 2023
Java	OpenJDK/Corretto 21
Maven	3.9.16
SonarQube Scanner CLI	7.3.0.5189

SonarQube	Docker Container
Nexus	Docker Container
Tomcat	Apache Tomcat
	9.0.122
Tomcat Port	8080
GitHub Branch	main
Application	Tomcat App01
Packaging	WAR

3. Application Repository

GitHub repository:

https://github.com/Zeeshancloud15/Tomcat_app01.git

Branch:

main

Application structure:

```
Tomcat_app01
├── pom.xml
├── sonar-project.properties
└── src
    ├── main
    │   └── webapp
    │       ├── index.jsp
    │       └── WEB-INF
    │           └── web.xml
```

4. Jenkins Server Setup

Jenkins was installed on an Amazon Linux 2023 EC2 instance.

Check Java

```
java -version
```

Java 21 was installed and verified.

Check Jenkins

```
sudo systemctl status jenkins
```

Start Jenkins if required:

```
sudo systemctl start jenkins
```

Enable Jenkins at boot:

```
sudo systemctl enable jenkins
```

Jenkins was accessed through:

<http://<JENKINS-PUBLIC-IP>:8080>

5. Maven Installation on Jenkins Server

Maven was installed directly on the Jenkins server.

Installation directory:

```
/opt/maven
```

Environment Configuration

Added to: vi .bash_profile

```
/root/.bash_profileexport MAVEN_HOME=/opt/maven  
export PATH=$MAVEN_HOME/bin:$PATH
```

Reload:

```
source /root/.bash_profile
```

Verify:

```
mvn -version
```

Expected:

Apache Maven 3.9.16

Jenkins Maven Configuration

Go to:

Jenkins

→ Manage Jenkins

→ Tools

→ Maven installations

Configuration:

Name: Maven-3.9.16

Maven Home: /opt/maven

Install automatically: Disabled

6. SonarQube Scanner CLI Installation

SonarQube itself is running in Docker, but the **SonarQube Scanner CLI** was installed directly on the Jenkins server.

Installation directory:

/opt/sonar-scanner

Environment Configuration

Added to:

```
/root/.bash_profileexport SONAR_SCANNER_HOME=/opt/sonar-scanner
export PATH=$SONAR_SCANNER_HOME/bin:$PATH
```

Reload:

```
source /root/.bash_profile
```

Verify:

```
sonar-scanner --version
```

Scanner version:

7.3.0.5189

7. SonarQube Installation Using Docker

SonarQube was deployed as a Docker container.

Check Docker

```
docker --version
```

Check Running Containers

```
docker ps
```

The SonarQube container was verified as running.

SonarQube is accessible on:

<http://<SONARQUBE-IP>:9000>

In the Jenkins job, SonarQube was accessed using the server's reachable IP:

<http://172.31.1.105:9000>

8. SonarQube Project Configuration

The application contains:

```
sonar-project.properties
```

Configuration:

```
sonar.projectKey=tomcat-app01
sonar.projectName=Tomcat App01
sonar.projectVersion=1.0
sonar.sources=src
sonar.sourceEncoding=UTF-8
```

Purpose

SonarQube analyzes the application source code and identifies code quality issues.

9. Nexus Installation Using Docker

Nexus Repository Manager was deployed as a Docker container.

Check Nexus Container

```
docker ps
```

Nexus was verified as running.

Nexus was accessible through:

<http://52.90.236.162:8081>

Nexus Repository

A Maven hosted repository was configured:

```
maven-release
```

This repository is used to store the generated WAR artifact.

10. Nexus Repository Configuration

Artifact details:

GroupId: com.zeeshan
ArtifactId: tomcat-app01
Version: 1.0
Type: war
Repository: maven-release

Expected artifact:

com/zeeshan/tomcat-app01/1.0/tomcat-app01-1.0.war

11. Tomcat Installation

Apache Tomcat 9.0.122 was installed on a separate EC2 server.

Installation directory:

/opt/tomcat

Tomcat port:

8080

Start Tomcat:

```
cd /opt/tomcat/bin  
./startup.sh
```

Check port:

```
ss -lntp | grep 8080
```

Tomcat was verified successfully.

12. Configure Tomcat Manager

Jenkins deploys the WAR using the Tomcat Manager interface.

Edit:

```
vi /opt/tomcat/conf/tomcat-users.xml
```

Configured:

```
<role rolename="manager-script"/>
<user username="jenkins"
      password="Jenkins@123"
      roles="manager-script"/>
```

13. Allow Jenkins Server to Access Tomcat Manager

Edit:

```
vi /opt/tomcat/webapps/manager/META-INF/context.xml
```

Configured Jenkins private IP:

```
<Valve className="org.apache.catalina.valves.RemoteCIDRValve"
      allow="127.0.0.0/8,::1/128,172.31.2.13" />
```

Here:

172.31.2.13

is the Jenkins server private IP.

Restart Tomcat:

```
cd /opt/tomcat/bin
./shutdown.sh
./startup.sh
```

14. Test Tomcat Manager Connectivity

From the Jenkins server:

```
curl -u 'jenkins:Jenkins@123' \  
http://172.31.11.181:8080/manager/text/list
```

Successful response:

```
OK - Listed applications for virtual host [localhost]
```

This confirmed that Jenkins could access Tomcat Manager.

15. Jenkins Tomcat Credentials

Created Jenkins credentials:

Kind: Username with password

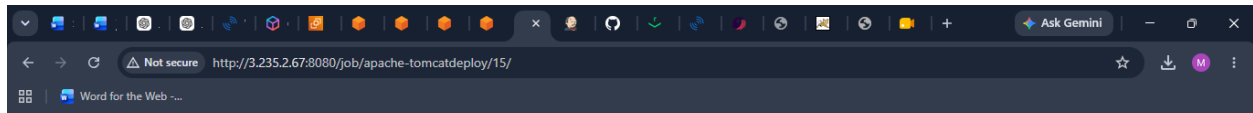
Username: jenkins

Password: Jenkins@123

ID: tomcat-jenkins

Description: Tomcat Manager

These credentials are used by Jenkins to deploy the WAR file to Tomcat.



Jenkins / apache-tomcatdeploy / #15

Status ✓ #15 (Oct 4, 2026, 12:57:33 PM) Add description Keep this build forever

Changes

Console Output

Delete build '#15'

Git Build Data

Edit build information

Previous Build

Next Build

Started by user [zeeshan](#)

Revision: a5124c72bc1aa536a78410c1138c9962cbc5b330

Repository: https://github.com/Zeeshancloud15/Tomcat_app01.git

- refs/remotes/origin/main

No changes.

Started 1 day 22 hr ago
Took 9.8 sec

REST API Jenkins 2.584



Git ?

Repositories ?

Repository URL ?

https://github.com/Zeeshancloud15/Tomcat_app01.git

Credentials ?

- none - + Add

Advanced ▾

JDK 

JDK to be used for this SonarQube analysis

(Inherit From Job) 

Path to project properties 

Analysis properties 

```
sonar.projectKey=tomcat-app01
sonar.projectName=Tomcat App01
sonar.projectVersion=1.0
sonar.sources=src
sonar.sourceEncoding=UTF-8
```

Additional arguments 

 Invoke top-level Maven targets 



Maven Version

maven-integration 

Goals

clean package 

Advanced 

Nexus Details

Nexus Version

NEXUS3

Protocol

HTTP

Nexus URL ?

52.90.236.162:8081

Credentials

admin/*****

+ Add

GroupId

WAR/EAR files ?

target/tomcat-app01.war

Context path ?

/zeeshan

Containers

Tomcat 9.x Remote



Credentials

jenkins/*****

+ Add

Tomcat URL ?

http://172.31.11.181:8080